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Reference Architecture and Agile Development Method for a Process-Driven Web Platform based on the BPMN-Standard and Process Engines

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Abstract

One of the major trends in the 21st century is the trend towards digital transformation within Industry 4.0. In order to adapt to high volatile market conditions, especially in the context of COVID-19, companies do not only have to transfer their knowledge into digital systems. Furthermore, they have to be able to tailor their business processes rapidly to changing customer's needs. The implementation of efficient and transparent processes is thereby known as the process of digital transformation and requires its underlying software-architecture to be highly flexible and scalable. Especially a web-based platform represents a highly user-centered application for multi-user purposes and can quickly adapt to its customer needs. While e-commerce and video-streaming platforms, like eBay or Netflix, often already provide a highly flexible, microservice-based architecture, common tools and platforms within the engineering-domain are usually still based on rather monolithic or service-oriented architectures (SOA). Therefore, an adaption to changing market conditions and the integration of proprietary tools is often technically demanding and economically not feasible, especially for small and medium-sized enterprises (SME). In this context, the so-called Process-Driven-Approach (PDA) offers a sustainable and tool-neutral blue-print for a system's architecture strictly following the separations-of-concerns principle and allowing a user-centered layout. The PDA enables the integration of business processes and tools within an orchestrating framework, provided by a process engine. While the PDA within the engineering domain has been presented in a preceding paper, this paper supposes a reference architecture and agile development method for engineering-platforms via the PDA based on the BPMN-standard and process engines. These concepts are validated by a prototypical implementation of a platform-demonstrator for tool-supported planning of robot-based automation solutions.

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Keywords: Process-Driven web platform; reference architecture; ROBOTOP; Process-Driven-Approach; PDA; agile; user-centered; development method

1. Introduction

The COVID-19 pandemic has had a heavy impact on the handling of day-to-day-business. As a result, most companies are having their employees work from home via different tools and web platforms. The sudden change of habits has stressed the previous existing urge to not only connect proprietary systems to web-based infrastructures, but also to flexibly adapt their business processes. Furthermore, the adaptability of processes in a larger context enables a faster reaction to

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changing customer's needs in volatile markets. Those requirements demand a highly flexible and scalable IT-architecture. Nevertheless, common systems are still based on monolithic or service-oriented architectures, complicating its adaptability [1–3]. Therefore, this paper shows a solution to flexibly combine business processes with IT-infrastructure, based on the Business Process Model and Notation (BPMN) standard [4] and introduces a reference architecture as well as a six-step method for its user-centered and agile development.

2. Relevant basics

During the past decades, digital transformation has yielded many innovations. While computer aided tools have simplified mass data handling, the coordination of individual processes still remains difficult due to point-to-point integration problems between different disciplines within planning, engineering, control and production [5]. For managing efficient and transparent engineering processes as well as an integrated tool chain, the Process-Driven Approach (PDA) offers a business-process-first methodology for creating sustainable, domain-oriented applications; potentially reducing adjustment costs.

2.1. BPMN-standard within the context of process automation

To establish an explicit understanding for continuous process automation, it is necessary to distinguish between non- and executable process models (before/ after BPMN) based on generic goals for using business process models, roles, as well as modeling and implementation approaches. Relevant goals can be differenciated in four general categories, with each having different responsibilities (Fig. 1). BPMN suggests itself for being used throughout as a lingua franca, from process modelling to seamless integration into a process engine. [6]

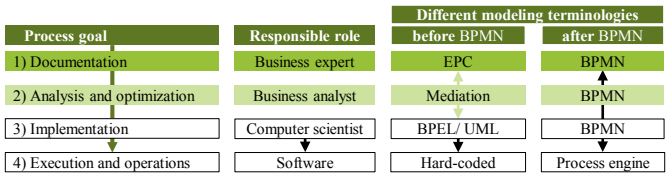


Fig. 1. Impact of BPMN: goals, roles and modeling terminologies [6]

Thereby, a process engine handles the process model execution, as well as orchestration, persisting, process data, reporting and overall management of tasks. (Fig. 2) [7].

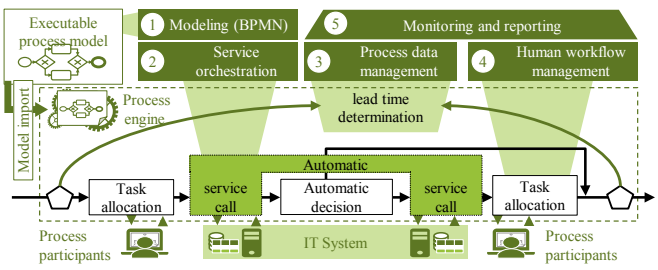


Fig. 2. Process model execution including phases (1-5) according to [6,7]

Intuitively, one would start from the business process and refine it top-down until reaching the system level and directly calling the back-end systems from the BPMN models. This approach reveals several disadvantages: The models become hardly understandable for non-BPMN-experts, are hard to maintain, adjust, and operate and even worse, get an increasingly monolithic structure. Therefore, the PDA offers a sustainable and tool-neutral blue-print for a system's architecture strictly following the separations-of-concerns principle.

2.2. Process-Driven Approach

Previously, a three-layered architecture for PDA usage within the engineering domain, based on the separation of concerns principle, was introduced [6]: The PDA-Layer (PDA-L), the Service Contract Implementation-Layer (SCI-L) and the Back-End-Layer (B-L) (see Fig. 3). A user-specific Front-end User Interface (F-UI) is used for human-machine interaction.

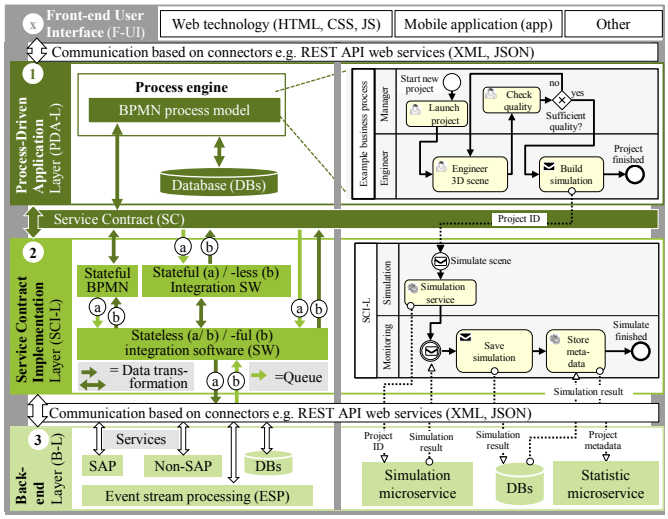


Fig. 3. Three-layered PDA architecture with human-machine interface [6]

The motivation for the division into layers is described in [6] and [8] in further detail. This allows a transparent and explicit separation of the business processes from the respective service implementations provided by the existing IT-landscape. Based on service contracts with input and output derived from the business processes' needs, the SCI-L therefore mediates between the PDA-L and legacy software-specific applications, systems or (micro-)services on the B-L. To make the PDA more accessible for the engineering domain, a minimally functional PDA architecture (MFPA) and its components have been composed in paper [6]. Hereby, an extendable framework with fundamental PDA functionalities is provided.

2.3. User-centered design

User-centered design (UCD) is an iterative process of system development, with the aim to identify and fulfil user's requirements in the best possible manner (see Fig. 4). [9]

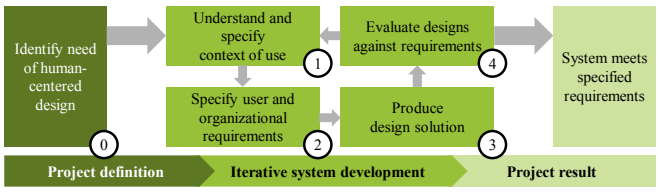


Fig. 4. User-centered design [9]

A common method for early phases 0-2 is the definition of exemplary personas, scenarios and use cases, which result in user-specifications regarding a system's usability, workflows and user-interfaces. Usability can be used to measure a system's and intermediate prototype's degree of requirement fulfilment, by evaluating its efficiency, effectiveness and satisfaction [9]. Nevertheless, since a developer usually will not be the typical user, it is essential to frequently observe and involve end-users throughout the entire development process. This enables an enormous potential of complexity reduction for the user and the developer. [10] The procedure of UCD can be combined with common agile methods, e. g. Scrum. Hereby, actions within steps 1-4 can be prioritized and handled within planned sprints and lead to incremental improvements. [11]

3. Need for action and research methodology

Heterogeneous systems, data of different domains (Fig. 5), lack of process automation and tracking lead to time and information losses in the engineering domain as well as to increased costs due to redundancies, errors and high management overheads.

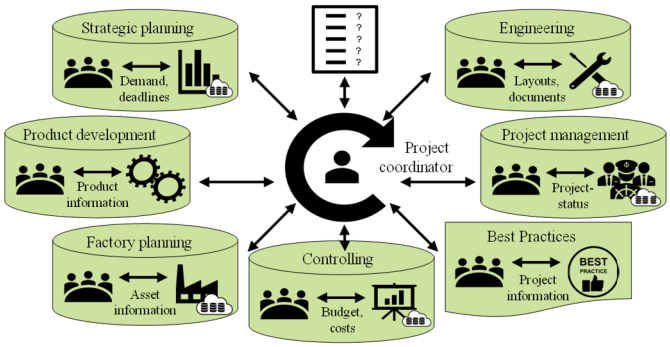


Fig. 5. Heterogeneous systems and data using the example of engineering

To reduce synchronization and communication overhead, a sustainable Process-Driven platform-infrastructure is needed.

In a previous research a MFPA for the engineering domain as well as an engineering-configurator was introduced [6,12]. The reference architecture and method for agile development of user-centered engineering platforms presented within this paper builds upon this MFPA and extending literature research.

In order to continuously improve the referenced content, multiple development cycles as well as user demonstrations and expert interviews have been performed. Based on aggregation of common literature knowledge and insights of different use cases within the ROBOTOP research project this paper proposes a reference architecture as well as a development method as shown in chapter 4 and 5. In section 6, a minimal functional architecture of a web platform within the engineering domain is evaluated using a prototypical implementation. Through interviews and literature research, improved understanding and potential evaluation was targeted.

4. Reference architecture for engineering-platforms

Since the PDA has successfully supported the digital transformation, including project management, implementation of process sequences as well as its overall architecture [13], it is now being adopted within the development of flexible and scalable engineering-platforms.

4.1. General concept for a Process-Driven web platform

The goal of the concept is to realize digital, cross-project planning consistency, with simultaneous transparent and comprehensible process coordination, automation, adaption and optimization. Therefore, it is necessary that the individualized information, task lists and tasks are visualized for the decision-makers, the project management and the respective users via suitable user interfaces. Especially in the context of heterogeneous and highly specialized software landscapes, it is necessary to integrate all existing software tools, whether CAx or PLM/PDM, as far as possible.

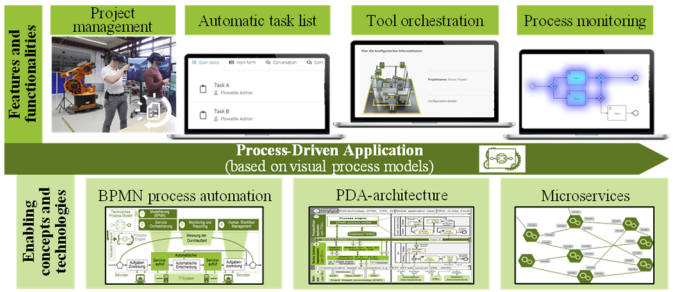


Fig. 6. General concept of a Process-Driven web platform

The focus is a scalable PDA architecture for management and engineering as well as the process modelling of continuous management and engineering processes based on BPMN (Fig. 6). The processes are to be executed prototypically by means of a suitable BPMN process engine. With the help of transparent processes and standardized interfaces, different, highly specialized management and engineering tools as well as microservices and apps can be consistently and continuously connected, controlled and replaced. The scalability of the solution and distributed data storage in the sense of big data can be guaranteed via a hybrid cloud solution, with simultaneous aggregation of domain expertise. This can be realized by means of cross-tool data aggregation and analysis services.

4.1. Requirements for modern web platform infrastructure

According to CHOUDARY [14] and TAEIHAGH [15], platform infrastructures enable crowdsourcing as well as collaborative consumption and generally function across three layers. Each layer provides certain functionalities and therefore has different requirements:

The **Network-Marketplace-Community layer** comprises the users and their interaction with each other. Therefore, a user and identity management as well as a communication system has to be provided (e. g. OAuth2/ OpenID Connect). Furthermore, front-ends for different user devices have to be designed and load balancing for optimizing response time has to be considered. The **Infrastructure layer** enables the user's value creation by providing the platform's core logic and orchestrating the necessary tools and services. Lastly, the **Data layer** contains and provides various goods, services or content. Supporting (micro-)services for e. g. the exchange of information or data transformation as well as external communication build the platforms backbone.

4.2. Reference architecture for PDW within engineering

Fig. 7 shows the reference architecture for a PDW, generally based on the PDA with respect to the layer-specific requirements mentioned in section 4.1.

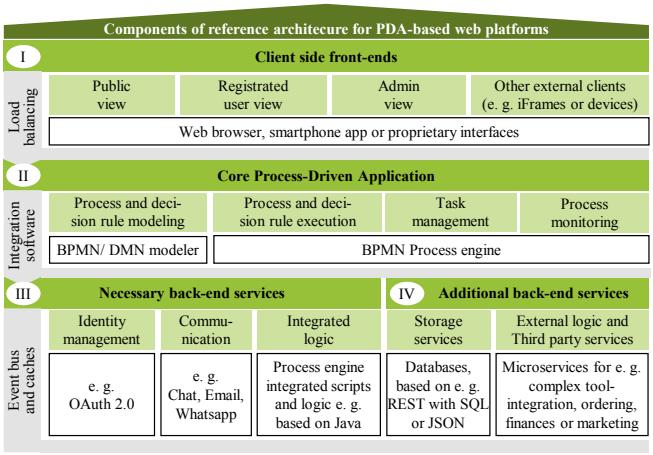


Fig. 7. Reference architecture for PDW [6,12]

I) Client side front-ends: For a simplified user interaction with the platform a suitable front-end visualization is needed. User interfaces running via web browsers offer a device-neutral front-end and can be implemented by using e. g. HTML, CSS or TypeScript-based frameworks like Vue.js or React. Smartphone apps and individual interfaces to other devices provide great mobility and "on-the-run" availability. The scope and amount of information shown to the user should be considered, depending on its status, role and accessibility rights. Load balancing can be realized by e. g. Ingress in a Kubernetes Hybrid Cloud or via NGINX.

II) Core process driven application: As depicted in section 2.1, the platform's control of the process logic is exclusively within the responsibility of the process engine. The connection to back-end services and databases should be handled via an asynchronous integration bus, as e. g. provided by Apache Kafka or RabbitMQ.

III) Necessary back-end services: Platform-basic components are handled within the back-end: Identity services for user and role management, communication services for platform-based messengers as well as further integrated scripts taking care of the technical back-end execution of processes and their logic.

IV) Additional back-end services: Depending on the use case, certain functionalities have to be outsourced to third party service providers. For instance, Google Maps API can be used for location services or SendGrid API for external communication. Furthermore, SQL or JSON-based databases (like MongoDB) provide storage capabilities for business data objects via e. g. REST. For better scalability, data exchange should be realized by making use of the integration bus, e. g. by integration patterns as provided by Apache Camel.

5. Agile and user-centered development method

For agile development of a scalable, sustainable and user-centered PDW, a six-step method based on ISO 9241-210 [9], ZIMMERMANN AND GRÖTZBACH [16] as well as insights from previous research [17] is now being introduced (Fig. 8). While step 1-3 set up the conceptual business case requirements, steps 4-6 concern the process design and the implementation of iteratively increasing detail. Scrum is recommended to be used for incremental improvements within the latter steps [11].

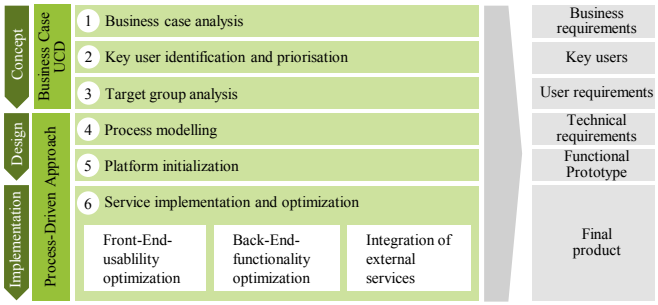


Fig. 8. Agile development method for user centered PDW

5.1. Business case and value creation network analysis

Within the first step the potential business cases should be analyzed, considering stakeholders, data, product and payment flows. This results in the business requirements and sets the overall scope of the platform, as exemplary shown in Fig. 9.

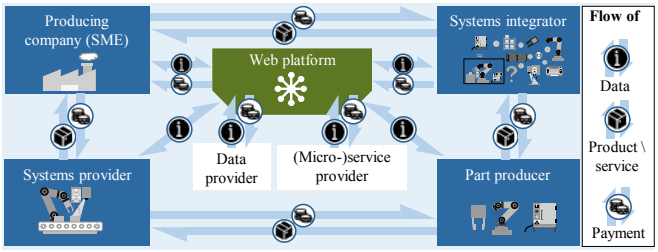


Fig. 9. Potential value network of a robotics web platform [12]

5.2. Key user identification and prioritization

Within customer segments, the key user groups have to be identified, e. g. by creating a business model canvas. In case multiple ones can be identified, a prioritization e. g. by using weighted decision matrix is recommended. Therefore, most of all, value proposition, key activities, revenue streams and channels should be considered, as illustrated in Fig. 10 [18].

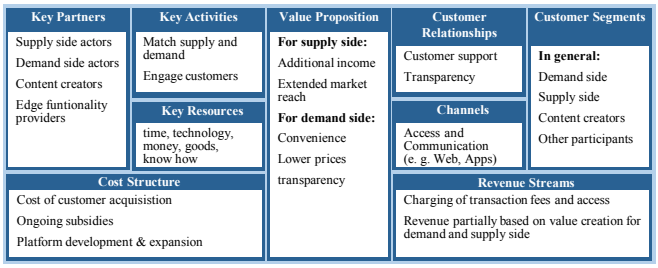


Fig. 10. Generic business model canvas of a web platform [18]

5.3. Target group analysis

Next, the identified target group has to be analyzed regarding their preferences, behaviors and needs. The goal is to determine certain personas and user scenarios, which can be carried on into usability, user and UI requirements [16]. Requirements concerning usability can be defined and evaluated by determining the effectiveness, efficiency and satisfaction needed. Information can be gathered by literature- and market-research especially using user-surveys. Thus, intense potential-user involvement at this point is essential.

5.4. Process modeling based on five steps of development

Based on SILVER [19], processes modeled in BPMN should be refined top-down in five steps with increasing process logic detail: Firstly, the overall scope and function of the general business process from start to different possible endings should be agreed on. In step 2, the major activities should be enumerated and described in a high-level list of activities with well-defined inputs and outcomes. Next, the top-level BPMN diagram should be modeled using (swim) pools and lanes, including the high-level activities as individual sub-processes or *Send Tasks* taken care of later. Besides the "happy path", possible exception paths should be considered as well. In step 4, each sub-process should be hierarchically extended within a child-level diagram. Finally, business context should be visualized by message flows between different (sub-) processes and service providers drawn as black boxes. Steps 4 and 5 can be repeated with additional nested levels, if needed.

5.5. Web platform initialization via minimal viable product

In order to get a functional prototype, the basic architecture from section 4.1 as well as the most urgent services should be implemented. For this, a detailed breakdown of the processes of the respective domain in order to derive the required services and UIs must be accomplished, as in previous research [17]. Some process engines (e. g. Flowable) with their enterprise versions offer "all-inclusive" packages for process modelling and execution, user and task management as well as complex front-end design. This enables rapid prototyping to be closer to the final product implementation.

5.6. (Micro-)service implementation and optimization

Iterative improvement to finish the product: Front-end-usability and visual design should be optimized until living up to the defined user's requirements. Back-end microservices should be further implemented and optimized for functionality and performance, as well as external, third-party services integrated if needed.

6. Validation

For validation purposes, a simple use case within the ROBOTOP research project has been selected [20]. While the development method's conceptual steps 1-3 have been processed in previous research [12], the validation described in this paper mainly focuses on the implementation of a flexible platform architecture. Firstly, the exemplified use case as well as its implementation will be described and secondly, general results from user and expert feedback cycles will be presented.

6.1. Validation through prototypical implementation

The exemplified implementation targets the establishment of a PDW matching demand and supply of consulting services. The simplified processes are shown in Fig. 11. The back-end integration has been realized with Apache Camel and messaging using the open source message broker Apache ActiveMQ Classic.

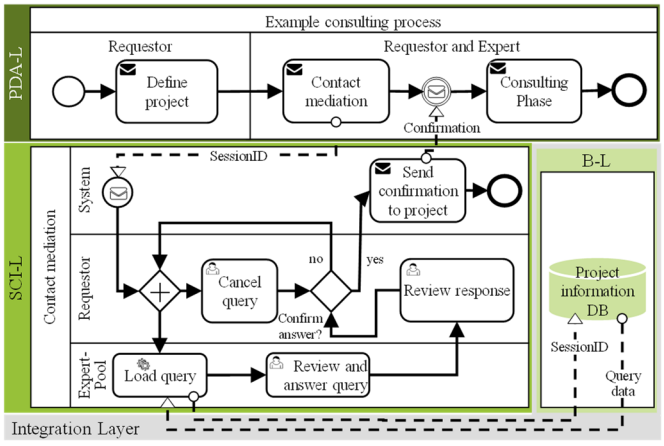


Fig. 11. Exemplified contact mediation process based on the PDA

Fig. 12 shows the front-end visualization as seen by a user with a consulting function. The user interface was designed to have an intuitive overview over still unanswered inquiries within a project feed and fast access to own projects.

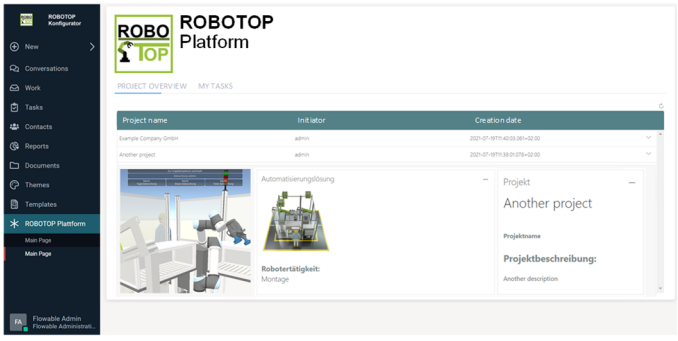


Fig. 12. Front-end demonstration of exemplified use case implementation

6.2. Validation through expert interviews and user tests

To qualitatively validate the results, the architecture, development method and prototype were presented to and discussed with several stakeholders. In this context, enterprise software development, business process automation and engineering experts were queried within a two-day workshop. In general, it was seen as a very promising approach, as graphical process models can increase agility as well as a better separation of concerns in development. The potential for the engineering domain was considered high due to the possible coordination and combination of various existing tools. The importance of a separated integration layer with e. g. Camel and ActiveMQ was particularly emphasised. Using user feedback, the flexibility of the architecture and agility of the method have been proven. Thus, several times within the framework of short-cyclic user feedback-based development loops, the processes could be redesigned and implemented, without any major additional effort compared to classic, non-process driven development approaches.

7. Conclusion and outlook on future research activities

Building upon previous research of the Process-Driven Approach (PDA) within the engineering domain, a reference architecture and agile development method for a Process-Driven web platform (PDWP) based on the BPMN-standard and process engines has been introduced. Based on the PDA principles, the process orchestration and execution of the PDWP is conducted via process engine. Thus, the referenced architecture features a three-layered structure, consisting of the front-end layer, the core process components including the process engine, and back-end (micro-)service implementations. The introduced agile development method is achieved through the direct process coupling of modelled and executed processes in combination with user-centered front-end development. For validation, a Process-Driven platform was implemented in the context of the research project ROBOTOP.

Future research within the PDA-RobE research project aims to develop a process-oriented digital twin for plant engineering that stores, consolidates and makes available project-relevant information about historical and current production lines as well as the associated equipment along the Product Development Process (see Fig. 13). This should significantly reduce the planning effort of new projects as well as enable the reusability of already existing equipment and the expertise acquired through it.

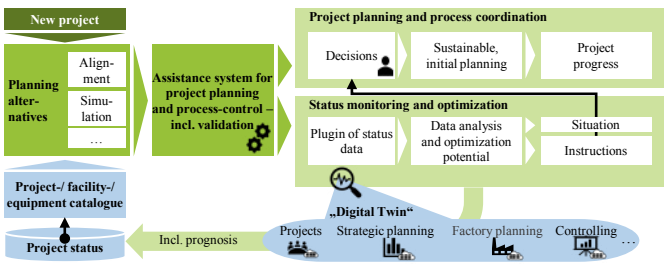


Fig. 13. Process loop for platform-supported and automated planning, orchestration, control and monitoring of engineering projects

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